

MATLAB Example: Calculating a Complex Function Using Random Inputs

Purpose of This Example

In this example, we use MATLAB to calculate the value of a mathematical expression for z using two input variables, x and y .

The values of x and y are randomly generated by MATLAB. Then, these values are used in a complex equation to calculate z .

This example helps you learn how to:

1. Generate random numbers in MATLAB
2. Use mathematical operations such as powers, square roots, absolute values, sine, and cosine
3. Break a long equation into multiple lines using `...`
4. Print results in a clear table format using `fprintf`

MATLAB Code

```
% Author: Prof. Lund
% Purpose: To solve a complex function for z given x and y.

clear all
clc

% Formatting some helpful output statements
fprintf('\n%5s%5s%10s\n', 'x','y','z');
fprintf('%5s%5s%10s\n', '----','----','----');

% Generating Random Input Variables
x = rand*15;
y = rand*15;

% Calculating the Output
z = (x^2-2*y)^3/(100*x) + ...
    abs(x*y)/(5*sqrt(10*x+y^(7/3))) + ...
    sin(x)*cos(y)/5;
% Notice here that the ... operator allows us to continue our operation
% on a new line! This can help clean up your code.

% Printing out the results
fprintf('%5.1f%5.1f%10.2f\n', x,y,z)
```

Detailed Explanation

1. Clearing the MATLAB Workspace

```
clear all  
clc
```

The command `clear all` removes all variables from the MATLAB workspace.

The command `clc` clears the Command Window.

These two commands help us start with a clean MATLAB environment before running the code.

2. Printing the Table Header

```
fprintf('\n%5s%5s%10s\n', 'x', 'y', 'z');  
fprintf('%5s%5s%10s\n', '----', '----', '----');
```

The function `fprintf` is used to print formatted text in the Command Window.

This part creates a table header:

```
      x      y      z  
----  ----  ----
```

Here:

```
%5s
```

means MATLAB prints a string using 5 spaces of width.

```
%10s
```

means MATLAB prints a string using 10 spaces of width.

The purpose is to make the output look organized.

3. Generating Random Values for x and y

```
x = rand*15;  
y = rand*15;
```

The function `rand` generates a random number between 0 and 1.

For example:

```
rand
```

may give a value such as:

```
0.4321
```

When we multiply it by 15:

```
rand*15
```

the result will be a random number between 0 and 15.

Therefore:

```
x = rand*15;  
y = rand*15;
```

creates two random input values:

```
0 < x < 15  
0 < y < 15
```

Each time you run the code, MATLAB will probably generate different values for x and y .

4. Calculating the Output z

```
z = (x^2-2*y)^3/(100*x) + ...  
    abs(x*y)/(5*sqrt(10*x+y^(7/3))) + ...  
    sin(x)*cos(y)/5;
```

This equation calculates z using the values of x and y .

The equation has three main parts.

Part 1

```
(x^2-2*y)^3/(100*x)
```

This part means:

1. Square x :

$$x^2$$

2. Multiply y by 2:

$$2*y$$

3. Subtract $2*y$ from x^2 :

$$x^2 - 2*y$$

4. Raise the result to the power of 3:

$$(x^2 - 2*y)^3$$

5. Divide by $100*x$:

$$(x^2 - 2*y)^3 / (100*x)$$

Part 2

$$\text{abs}(x*y) / (5*\text{sqrt}(10*x+y^{(7/3)}))$$

This part contains the absolute value and square root.

First, MATLAB calculates:

$$x*y$$

Then:

$$\text{abs}(x*y)$$

The function `abs()` gives the absolute value.

For example:

$$\text{abs}(-8)$$

gives:

$$8$$

Then MATLAB calculates the denominator:

```
5*sqrt(10*x+y^(7/3))
```

Here:

```
sqrt(...)
```

calculates the square root.

Also:

```
y^(7/3)
```

means y raised to the power of $7/3$.

So the second part is:

```
abs(x*y)/(5*sqrt(10*x+y^(7/3)))
```

Part 3

```
sin(x)*cos(y)/5
```

This part uses trigonometric functions.

```
sin(x)
```

calculates the sine of x .

```
cos(y)
```

calculates the cosine of y .

Important note: In MATLAB, `sin()` and `cos()` use radians, not degrees.

So, x and y are treated as radian values.

5. Using the Three Dots Operator ...

In the equation, we have:

```
z = (x^2-2*y)^3/(100*x) + ...
    abs(x*y)/(5*sqrt(10*x+y^(7/3))) + ...
    sin(x)*cos(y)/5;
```

The symbol:

...

is called the continuation operator.

It tells MATLAB that the command continues on the next line.

Without ..., MATLAB may think the command is finished at the end of the line.

Using ... makes long equations easier to read.

For example, instead of writing one very long line:

```
z = (x^2-2*y)^3/(100*x) + abs(x*y)/(5*sqrt(10*x+y^(7/3))) + sin(x)*cos(y)/5;
```

we can write it in multiple lines:

```
z = (x^2-2*y)^3/(100*x) + ...  
    abs(x*y)/(5*sqrt(10*x+y^(7/3))) + ...  
    sin(x)*cos(y)/5;
```

This is cleaner and easier to understand.

6. Printing the Final Result

```
fprintf('%5.1f%5.1f%10.2f\n', x,y,z)
```

This line prints the final values of x , y , and z .

Here:

```
%5.1f
```

means MATLAB prints a decimal number with:

- total width of 5 spaces
- 1 digit after the decimal point

So, if:

```
x = 8.2345
```

MATLAB prints:

8.2

Also:

```
%10.2f
```

means MATLAB prints a decimal number with:

- total width of 10 spaces
- 2 digits after the decimal point

So, if:

```
z = 123.4567
```

MATLAB prints:

```
123.46
```

Example Output

Because `x` and `y` are random, your output may be different each time.

An example output may look like this:

```

      x      y      z
---- ----
 8.4 12.1 47.25
```

This means MATLAB generated:

```

x = 8.4
y = 12.1
z = 47.25
```

Your values may be different because `rand` generates new random numbers each time.

Important Notes for Students

1. The command `rand*15` generates a random number between 0 and 15.
2. The symbol `^` is used for powers in MATLAB.

Example:

x^2

means x squared.

3. The function `sqrt()` calculates the square root.

Example:

```
sqrt(25)
```

gives:

5

4. The function `abs()` calculates the absolute value.

Example:

```
abs(-10)
```

gives:

10

5. The functions `sin()` and `cos()` use radians in MATLAB.
6. The `...` operator is useful for writing long equations on multiple lines.
7. The function `fprintf()` is useful for printing organized output in the Command Window.

Practice Question for Students

Modify the code so that x and y are not random numbers.

Use:

```
x = 10;  
y = 5;
```

Then calculate and print the value of z .

Practice Code

```

clear all
clc

fprintf('\n%5s%5s%10s\n', 'x','y','z');
fprintf('%5s%5s%10s\n', '----','----','----');

x = 10;
y = 5;

z = (x^2-2*y)^3/(100*x) + ...
    abs(x*y)/(5*sqrt(10*x+y^(7/3))) + ...
    sin(x)*cos(y)/5;

fprintf('%5.1f%5.1f%10.2f\n', x,y,z)

```

Expected Result

For:

```

x = 10;
y = 5;

```

the output will be approximately:

x	y	z
----	----	----
10.0	5.0	729.08

This means that when $x = 10$ and $y = 5$, the calculated value of z is approximately:

```

z = 729.08

```